

assignment-04-code

September 10, 2026

1 Assignment 04

1.1 Import External Libraries

```
[1]: import numpy as np
import sympy as sp

from IPython.display import display, Math
```

1.2 1. Find Eigenvalues using Colab

```
[2]: sigmax = np.array([[0, 1], [1, 0]])

display(Math(f"\sigma_x = {sp.latex(sp.Matrix(sigmax))}"))

print("-----")

eigenvalues, eigenvectors = np.linalg.eig(sigmax)
eigenvalues = np.round(eigenvalues, 4)
eigenvectors = np.round(eigenvectors, 4)

display(Math(f"\text{{Eigenvalues: }} {sp.latex(sp.Matrix(eigenvalues))}"))
display(Math(f"\text{{Eigenvectors: }} {sp.latex(sp.Matrix(eigenvectors))}"))
```

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Eigenvalues: $\begin{bmatrix} 1.0 \\ -1.0 \end{bmatrix}$

Eigenvectors: $\begin{bmatrix} 0.7071 & -0.7071 \\ 0.7071 & 0.7071 \end{bmatrix}$

1.3 4. Vector normalization by hand and using python 2

```
[3]: # --- Problem 4: check Problem 3 numerically ---
sigmax = np.array([[0, 1], [1, 0]])

# |>, |-> are the eigenvectors of sigma_x; np.linalg.eig returns them unit-normalized
_, eigvecs = np.linalg.eig(sigmax)
plus, minus = eigvecs[:, [0]], eigvecs[:, [1]]

display(Math(f"\ket{{+}} = {sp.latex(sp.Matrix(np.round(plus, 4)))}, \quad "
f"\ket{{-}} = {sp.latex(sp.Matrix(np.round(minus, 4)))}"))
```

```
# normalization check: <psi|psi> = 1 (vectors are real, so transpose = conjugate transpose)
for name, psi in [("+", plus), ("-", minus)]:
    print(f"<{name}|{name}> = {np.dot(np.transpose(psi), psi)[0, 0]:.4f}")

print("-----")

# spin up / down projectors from Problem 2: |0><0| and |1><1| (outer product via np.dot)
zero, one = np.array([[1], [0]]), np.array([[0], [1]])
P_up = np.dot(zero, np.transpose(zero))
P_down = np.dot(one, np.transpose(one))

# Born rule: P_psi(m) = <psi| P_m |psi>
for name, psi in [("+", plus), ("-", minus)]:
    p_up = np.dot(np.transpose(psi), np.dot(P_up, psi))[0, 0]
    p_down = np.dot(np.transpose(psi), np.dot(P_down, psi))[0, 0]
    display(Math(f"\text{{{P}}}_{{{name}}}(\uparrow) = {p_up:.4f}, \quad "
                f"\text{{{P}}}_{{{name}}}(\downarrow) = {p_down:.4f}"))
```

$$|+\rangle = \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}, \quad |-\rangle = \begin{bmatrix} -0.7071 \\ 0.7071 \end{bmatrix}$$

$$\langle + | + \rangle = 1.0000$$

$$\langle - | - \rangle = 1.0000$$

$$P_+(\uparrow) = 0.5000, \quad P_+(\downarrow) = 0.5000$$

$$P_-(\uparrow) = 0.5000, \quad P_-(\downarrow) = 0.5000$$

1.4 7. Finding the Determinants of Pauli Matrices using Colab

```
[4]: # --- Problem 7: determinants of the Pauli matrices ---
sigmax = np.array([[0, 1], [1, 0]])
sigmay = np.array([[0, -1j], [1j, 0]])
sigmaz = np.array([[1, 0], [0, -1]])

for name, M in [("\sigma_x", sigmax), ("\sigma_y", sigmay), ("\sigma_z", sigmaz)]:
    det = np.linalg.det(M)
    display(Math(f"\det {name} = {det.real:.4f}"))
```

$$\det \sigma_x = -1.0000$$

$$\det \sigma_y = -1.0000$$

$$\det \sigma_z = -1.0000$$